

Disagreement-Aware Probabilistic U-Net: Making Predictive Diversity Land Where Annotators Actually Disagree

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Selected paper. S. A. A. Kohl et al., *A Probabilistic U-Net for Segmentation of Ambiguous Images*, NeurIPS 2018 [1].

Proposed extension. We supervise *where* human annotators disagree. A single-convolution head predicts the inter-rater disagreement map from the image alone, and an alignment loss drives the model’s own sample diversity into those same pixels.

1 Introduction

Semantic segmentation assigns a label to every pixel of an image, and training normally assumes that each image has one correct mask. In medical imaging that assumption does not hold. Several qualified experts asked to outline the same lesion will produce visibly different masks, and on the LIDC-IDRI lung-nodule dataset they sometimes disagree about whether a lesion is present at all. A conventional U-Net trained on one expert’s mask, or on the average of several, returns a single confident segmentation and discards this information precisely where it is most clinically relevant.

The Probabilistic U-Net addresses this by combining a U-Net with a conditional variational autoencoder. A prior network maps the image x to a distribution $p(z | x)$ over a low-dimensional latent space; each draw of z is decoded into a different plausible segmentation. The samples are coherent whole masks rather than independent per-pixel probabilities, which is the paper’s central design argument: a pixel-wise probability map cannot express which pixels vary *together*, and so cannot represent two distinct but internally consistent interpretations of the same image. Training minimises the negative evidence lower bound, in which a posterior $q(z | x, Y_g)$ additionally observes one randomly chosen expert mask Y_g ,

$$L_{\text{PU-Net}} = \underbrace{\mathbb{E}_{z \sim q(z|x, Y_g)} [-\log p(Y_g | x, z)]}_{\text{reconstruction (cross-entropy, summed over pixels)}} + \beta \underbrace{D_{\text{KL}}(q(z | x, Y_g) \| p(z | x))}_{\text{pulls the prior towards the posterior}},$$

so that at test time sampling from the prior alone reproduces the variability seen across annotators. The balance between the two terms is consequential: Section 4 reports how an incorrect reduction of the reconstruction term collapses the posterior onto the prior and removes the model’s stochasticity entirely.

The resulting distribution of segmentations is scored with the generalized energy distance. Writing S, S' for independent model samples and Y, Y' for independent grader masks,

$$D_{\text{GED}}^2 = 2 \mathbb{E}[d(S, Y)] - \underbrace{\mathbb{E}[d(S, S')]}_{\text{sample diversity}} - \mathbb{E}[d(Y, Y')], \quad d = 1 - \text{IoU},$$

where $\text{IoU}(A, B) = |A \cap B| / |A \cup B|$ is the intersection over union of two binary masks.

We chose this paper because LIDC-IDRI provides four independent expert masks per image, which makes inter-rater disagreement directly measurable rather than merely assumed — a rare property among segmentation datasets. It also exposes a specific gap. Every term of D_{GED}^2 is a scalar: $\mathbb{E}[d(S, S')]$ measures how *much* the model’s samples vary and $\mathbb{E}[d(Y, Y')]$ how much the graders vary, but nothing in the metric asks *where* in the image that variation falls. A model could therefore produce the right amount of variability in entirely the wrong places and score just as well as one that does not. Answering the “where” question requires the two spatial maps that lie behind those scalars, and comparing them is what we add.

2 Proposed extension

Three quantities must be kept distinct. The first, D^{GT} , is *human* disagreement, computed from the four masks alone. With $P_i = \frac{1}{G} \sum_g Y_i^{(g)}$ the per-pixel mean over $G = 4$ graders, D^{GT} is its normalised binary entropy,

$$D_i^{\text{GT}} = -\frac{P_i \log(P_i + \epsilon) + (1 - P_i) \log(1 - P_i + \epsilon)}{\log 2} \in [0, 1],$$

which is 0 wherever all four graders agree and 1 where they split two against two. The second, D , is our model’s *prediction* of D^{GT} . The third, U , is *model* uncertainty: we draw K latents from the prior, decode each into a foreground-probability map $Q^{(k)}$, average them into $\bar{Q}_i = \frac{1}{K} \sum_k Q_i^{(k)}$, and apply the same functional,

$$U_i = -\frac{Q_i \log(Q_i + \epsilon) + (1 - Q_i) \log(1 - Q_i + \epsilon)}{\log 2}.$$

Because U and D^{GT} differ only in what is averaged — model samples in one case, grader votes in the other — any discrepancy between them is a discrepancy of location on a common $[0, 1]$ scale, which is what makes $|U - D^{\text{GT}}|$ a meaningful quantity to minimise.

Our extension adds two components, both deliberately minimal. The **disagreement head** is a single 1×1 convolution followed by a sigmoid, applied to the last U-Net feature map and tapped *before* the latent z is injected, so that $\hat{D}$ is a function of the image alone; it is trained with $L_D = \text{MSE}(\hat{D}, D^{\text{GT}})$. The **alignment loss** $L_{\text{align}} = \frac{1}{HW} \sum_i |U_i - D_i^{\text{GT}}|$ penalises predictive diversity that lands anywhere other than where the graders differ. The full objective is $L = L_{\text{PU-Net}} + \lambda_D L_D + \lambda_A L_{\text{align}}$, and Figure 1 shows both additions in place.

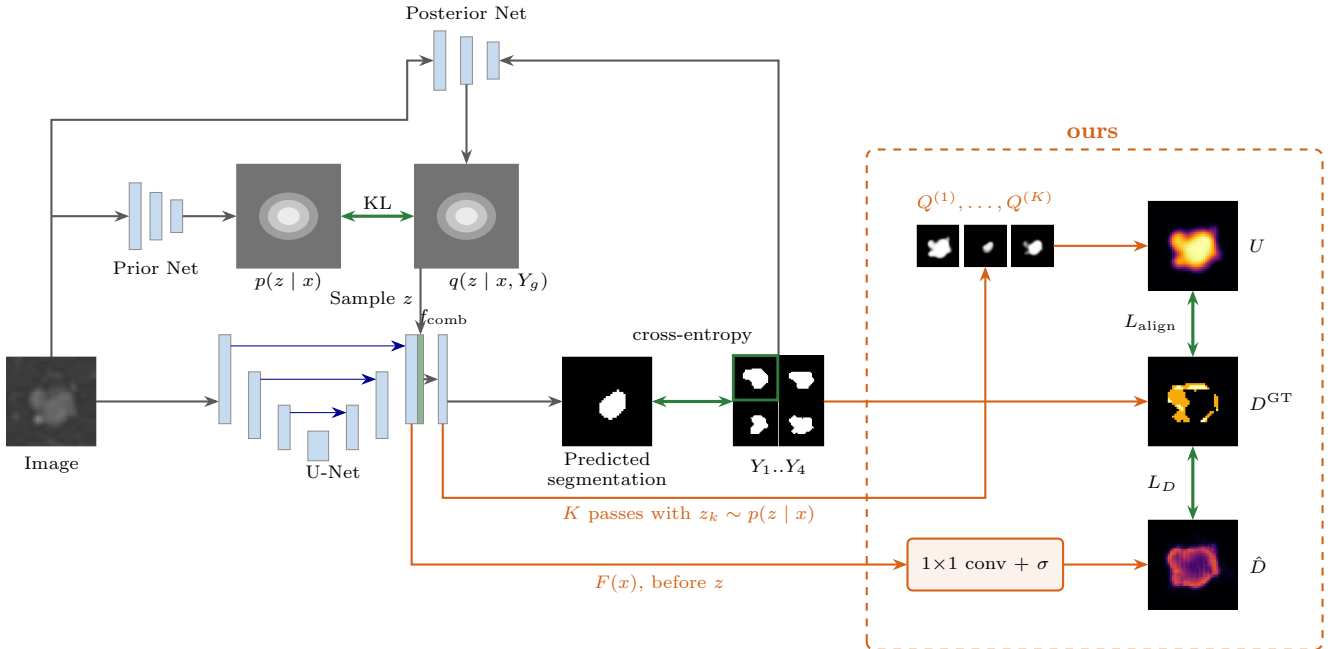


Figure 1: **The disagreement-aware Probabilistic U-Net (training).** Outside the dashed box, the original model: the posterior sees the image and *one* randomly drawn grader Y_g (green outline), and the sampled z (green block, 6 channels) is concatenated to the last U-Net activation (32 channels) and the three 1×1 convolutions of f_{comb} combine them; block width reads as channel count. Inside the box, our extension. Two things the drawing encodes: $F(x)$ is tapped *before* z , so $\hat{D}$ depends on the image alone; and the K passes behind U draw fresh z_k from the *prior*. **Every thumbnail is real**, from test crop LIDC-IDRI-0217 decoded by our trained full model and zoomed to the nodule; D^{GT} , U and $\hat{D}$ share one $[0, 1]$ colour scale, so they can be compared directly. The three $Q^{(k)}$ shown are non-empty draws — 8 of the 16 samples for this crop are empty, though all four graders segmented the nodule.

It is worth being precise about how these additions relate to the original design. D^{GT} and U are per-pixel marginals: projections of the joint distribution over masks that the paper deliberately models. We do not substitute

one object for the other. The conditional VAE continues to own the joint — the KL term keeps the prior broad and the decoded samples internally coherent — and L_{align} constrains only where the marginal variance is located. U is computed from the K decoded samples, so it is a readout of that distribution rather than a parallel pixel-wise model. Four masks determine the per-pixel Bernoulli exactly, so that with $G = 4$ the map D^{GT} takes only the values $\{0, 0.81, 1\}$, but they say very little about the joint; a marginal constraint is close to all that four annotations can support. The head $\hat{D}$ is genuinely pixel-wise, a 1×1 convolution on image features with no sample involved, and functions as an auxiliary output alongside the sample-based segmentation rather than as a replacement for it.

This yields two testable claims. **Q1:** where annotators will disagree is predictable from the image alone, so that $\hat{D}$ approximates D^{GT} and outperforms a trivial predictor that merely marks lesion boundaries. **Q2:** adding L_{align} increases the agreement between U and D^{GT} relative to the baseline without degrading segmentation quality or the energy distance.

We considered three alternatives. Conditioning the prior on disagreement, $p(z \mid x, \hat{D})$, was rejected because $\hat{D}$ is a deterministic function of x and therefore contributes no new information while confounding the ablation. Per-rater output heads or annotator confusion matrices [5] were rejected because the LIDC release does not identify which radiologist produced which mask and the four raters differ from scan to scan, leaving a per-rater parameterisation nothing stable to learn. The third alternative, replacing the ELBO with direct distribution matching, treats the four masks as an empirical distribution and trains the K samples to match it through a two-way soft-Dice loss. This is close in spirit to MoSE [6], which matches a segmentation distribution to an annotation set with a Wasserstein-like loss on the same dataset. Rather than dismiss it on paper, we implemented it and ran it as a fourth arm; Section 4 reports the outcome.

3 Methodology

Model. We reimplemented the Probabilistic U-Net in PyTorch, the official release being an older TensorFlow codebase that does not run on current Colab. The architecture follows the original: a U-Net backbone with 32 base channels and four down- and up-sampling stages, a prior encoder, a posterior encoder, a six-dimensional latent, and `fcomb`-style injection of z into the final 1×1 convolution stack. Our additions live in a subclass, so the unmodified baseline remains runnable and every arm shares one code path.

Data. We use the LIDC-IDRI 2D crops from DeepMind’s preprocessed release (CC BY 3.0), published with the Hierarchical Probabilistic U-Net [2], whose preprocessing follows Appendix H.1 of the original paper. The splits contain 8843 training, 1993 validation and 1980 test images from 530, 111 and 103 patients respectively; images are 180×180 8-bit grayscale with four masks each. We take a random 128×128 crop, scale images to $[0, 1]$ and binarise the masks. These counts are up to 0.6% below the 8882 / 1996 / 1992 reported in the paper, because the authors cleaned the dataset after publication and the 2018 snapshot was never released; the numbers here refer to the HPU-Net release. Roughly 65% of training crops contain at least one empty mask, reflecting disagreement about lesion *existence* rather than boundary placement, which is why the empty-mask convention of Section 4 matters.

Training. The objective, optimiser and schedule are taken unchanged from the original paper: the reconstruction cross-entropy summed over pixels, $\beta = 1$ with a six-dimensional latent, Adam with learning rate 10^{-4} decayed to 10^{-6} in five steps, batch size 32 and weight decay 10^{-5} . None of these were tuned, and every arm runs them identically, which is what allows the results table to be read as an ablation: a difference between arms cannot be an artefact of optimiser search, because no such search was performed.

Four choices were ours. We train for 100k steps rather than the paper’s 240k, a computational constraint; absolute energy distances consequently sit above published values, and only the relative ordering of arms trained to an identical budget is meaningful. We set $\lambda_D = 0.01$ and $\lambda_A = 10^{-5}$ after running short 10k-step trials across a range of values, which showed that the auxiliary terms buy little and cost a great deal: raising λ_A to 10^{-3} improved the uncertainty correlation by roughly 0.02 while Dice fell from 0.30 to 0.12, and 10^{-4} became unstable once the `fcomb` was corrected, so we kept the smallest weights that still had a measurable effect. We clip the gradient norm at 100 and clamp the latent log-variance to ± 10 , both required for numerical stability as described in Section 4. Augmentation is limited to the random translated crop; the paper additionally applies elastic deformation, rotation, shearing and scaling. Training draws $K = 4$ samples per step, and evaluation uses 16 samples per image.

Ablation design. Four arms vary one component at a time.

Arm	Disagreement head	Alignment loss	Objective
A – baseline PU-Net	no	no	$L_{\text{PU-Net}}$
B – + head	yes	no	$+\lambda_D L_D$
C – full model	yes	yes	$+\lambda_D L_D + \lambda_A L_{\text{align}}$
D – distribution matching	yes	no	$\lambda_{\text{dist}} L_{\text{dist}} + \lambda_{\text{con}} L_{\text{con}} + \lambda_D L_D$

Arm D discards the ELBO entirely: there is no random-grader cross-entropy, no posterior and no KL term. Instead K samples are drawn from the prior with gradients, and a $K \times G$ matrix of soft-Dice distances to the four masks is reduced in one of two ways. The *coverage* form requires every human mask to be matched by some sample and every sample to resemble some human mask, through a soft minimum; the *kernel* form uses a two-way soft kernel likelihood. L_{con} is a soft Dice between the mean sample and the mean human mask. Both modes are reported.

Tools. Python, PyTorch, NumPy and Matplotlib. The submitted code runs in Google Colab as a self-contained notebook that executes the whole pipeline end-to-end in a reduced “smoke” configuration; the same code paths, run at full length, produced every number reported below.

4 Results and analysis

Metrics. Dice and IoU, with $\text{Dice}(A, B) = 2|A \cap B|/(|A| + |B|)$, serve as a check that the extension does not damage segmentation quality; they are computed on the deterministic prediction against each of the four masks and are not the headline claim. The generalized energy distance, defined in Section 1, compares our 16 samples with the four human masks. Both it and Dice are evaluated under the empty-mask convention of the paper’s Appendix B: IoU and Dice are both 0/0 when two masks are empty, so the paper defines $d \equiv 0$ in that case, and agreement that no lesion is present is rewarded rather than penalised. We apply the corresponding rule to Dice and IoU by excluding empty-versus-empty pairs instead of scoring them zero; without it, the 65% of crops containing an empty mask depress every model equally and mask real differences. The correlation $\text{corr}(U, D^{\text{GT}})$ — whether the model becomes uncertain in the same places humans disagree — is the primary result.

Two further diagnostics are needed. Because disagreement in LIDC concentrates on lesion boundaries, a map that merely marks a boundary already correlates with D^{GT} , so $\hat{D}$ is compared against a trivial control: the 5-pixel boundary band (dilation minus erosion) of the model’s own predicted mask, extending roughly two pixels inside and two outside its edge. It uses no grader information and is therefore available at inference, unlike a band drawn around the ground truth. Finally, the subtracted $\mathbb{E}[d(S, S')]$ term of D_{GED}^2 is reported on its own, since it detects samples that have collapsed onto a single mask, alongside *nestedness*: the fraction of images for which, sorting that image’s samples by area, each is wholly contained in the next. Nestedness catches a subtler failure in which the model only inflates and shrinks one shape without ever proposing a different one, so that its samples can appear diverse while carrying a single degree of freedom.

Main results. Table 1 reports the held-out test split (1980 images, 16 samples per image), with every arm read from its matched 100k-step checkpoint.

Arm	Dice $\uparrow$	IoU $\uparrow$	GED $\downarrow$	$\text{corr}(U, D^{\text{GT}}) \uparrow$	$\text{corr}(\hat{D}, D^{\text{GT}}) \uparrow$	control	$\mathbb{E}[d(S, S')]$	nested $\downarrow$
A – baseline	0.355	0.280	0.394	0.483	—	0.417	0.686	1%
B – + head	0.308	0.241	0.403	0.492	0.609	0.451	0.687	3%
C – full	0.340	0.267	0.388	0.509	0.627	0.425	0.692	1%
D – coverage	<i>0.438</i>	<i>0.369</i>	0.714	0.120	0.565	0.507	0.024	99.5%
D – kernel	<i>0.405</i>	<i>0.337</i>	0.933	0.103	0.569	0.426	0.000	100%

Table 1: Test-set results. *control* is the boundary-band predictor for the same model, so $\text{corr}(\hat{D}, D^{\text{GT}})$ should be read against the value beside it. Dice for arm D is italicised because it is not comparable; see the discussion below.

Q1 is confirmed. The head reaches $\text{corr}(\hat{D}, D^{\text{GT}}) = 0.627$ against its own boundary-band control of 0.425, a margin of +0.20. It has therefore learned something about annotator behaviour rather than simply tracing the

outline of its own prediction, and the result holds in every arm carrying a head. This is the clearest positive finding of the project.

Q2 is supported but modest. The correlation $\text{corr}(U, D^{\text{GT}})$ rises from 0.483 to 0.509 between arms A and C, and arm C simultaneously attains the best energy distance of the three ELBO arms (0.388) at a relative Dice cost of 4%. We read the small size of the gain as structural rather than as a tuning failure. L_{align} constrains a marginal of a joint object, so two sample distributions with identical marginals are indistinguishable to it: it can relocate variance but cannot reshape modes. The effect only became measurable at all once the latent was able to vary spatially. Our **fcomb** initially lacked a nonlinearity between its 1×1 convolutions and therefore collapsed to a single affine map; because the injected z is spatially constant, every sample became a level set of one function and 100% of test images had strictly nested samples, leaving U as a single ring that no loss could reshape. With the activation applied, nestedness falls to the 1–3% of Table 1.

Arm D wins on Dice and loses on everything else. Direct distribution matching produces the best Dice by a wide margin (0.438 against 0.355) and the worst energy distance by a wide margin (0.714 and 0.933 against 0.388). The mechanism is visible in a single column: $E[d(S, S')]$ is 0.024 for the coverage form and 0.0004 for the kernel form, against 0.69 for arms A–C, with 99–100% of test images nested. All sixteen samples are effectively the same mask, so the model has become deterministic and the subtracted diversity term of D_{GED}^2 vanishes, which is exactly why the energy distance nearly doubles even as Dice improves. The cause is that a two-way soft-Dice objective combined with a consensus term is minimised by placing every sample on the single mask closest to all four annotations; nothing in it rewards spread. In the original ELBO the KL term is what keeps the prior broad, and removing the conditional VAE removed the only pressure holding the samples apart. The Dice advantage is therefore circular: arm D is trained on a soft-Dice criterion and then evaluated by Dice on what has become a point estimate. It is a better point estimate and a considerably worse model of ambiguity. Arm D is in this sense the mirror image of our own extension: it attempts to obtain the joint distribution from the loss, having deleted the machinery that maintained it, whereas we retain that machinery and constrain only its marginal.

Arm D also illustrates why correlation rather than error magnitude is the right primary metric. It attains the *lowest* disagreement MAE of any arm (0.0125 against 0.0176 for arm C), because D^{GT} is near zero almost everywhere and a model predicting no uncertainty anywhere scores well on a mean absolute error. Correlation exposes this immediately; MAE does not.

Failure cases and challenges. Our first full-length run suffered posterior collapse and was discarded. The reconstruction term averaged cross-entropy over pixels where Kohl et al. sum it; at 128×128 this under-weights reconstruction by a factor of about 16000 relative to β KL, so $\beta = 1$ behaved like $\beta \approx 16000$. The KL term reached zero by step 31k and the validation energy distance *rose* from 0.34 to 0.61 while Dice continued to improve — the signature of a network that had become deterministic. Summing the cross-entropy fixed it. With the corrected loss, gradients are far larger and every arm produced NaNs in the prior encoder within 2k steps, which is why the clipping and clamp of Section 3 are required. A related instability appears at $\lambda_A = 10^{-4}$, where the KL doubles at every step until it overflows: L_{align} back-propagates into the prior, and inflating the prior variance is the cheapest available way to raise sample diversity. Detaching the prior in the sampling path removes this route.

Checkpoint selection also proved misleading. Selecting on best validation energy distance chooses step 1000 for arm A and step 3000 for arm B, because a barely-trained model with a wide, uninformative prior scores well on that metric; we therefore report the matched 100k-step checkpoint for every arm, which compares methods rather than training durations. Finally, Figure 2 shows a conceptual limit. Much of the disagreement in LIDC concerns lesion existence, so matching U to D^{GT} demands maximal uncertainty across whole lesions, in direct opposition to the reconstruction term. This bounds how far L_{align} can usefully go and motivates the refinement proposed below.

5 Conclusion

Where radiologists will disagree is predictable from the image alone: a single 1×1 convolution reaches a correlation of 0.627 with the true disagreement map, against 0.425 for a trivial boundary-band predictor. Aligning sample diversity with that disagreement produces a smaller but consistent gain, raising $\text{corr}(U, D^{\text{GT}})$ by 0.026 while attaining the best energy distance among our ELBO arms at a 4% Dice cost. The claim we make is not that our Dice is higher. The baseline already generates diverse plausible segmentations; our extension makes that diversity correspond to where experts actually disagree, and turns the disagreement itself into an explicit, inspectable output. Summarised in one sentence, the design takes its joint structure from the conditional VAE and its marginal localisation from our loss, and arm D demonstrates what happens when the joint is instead demanded

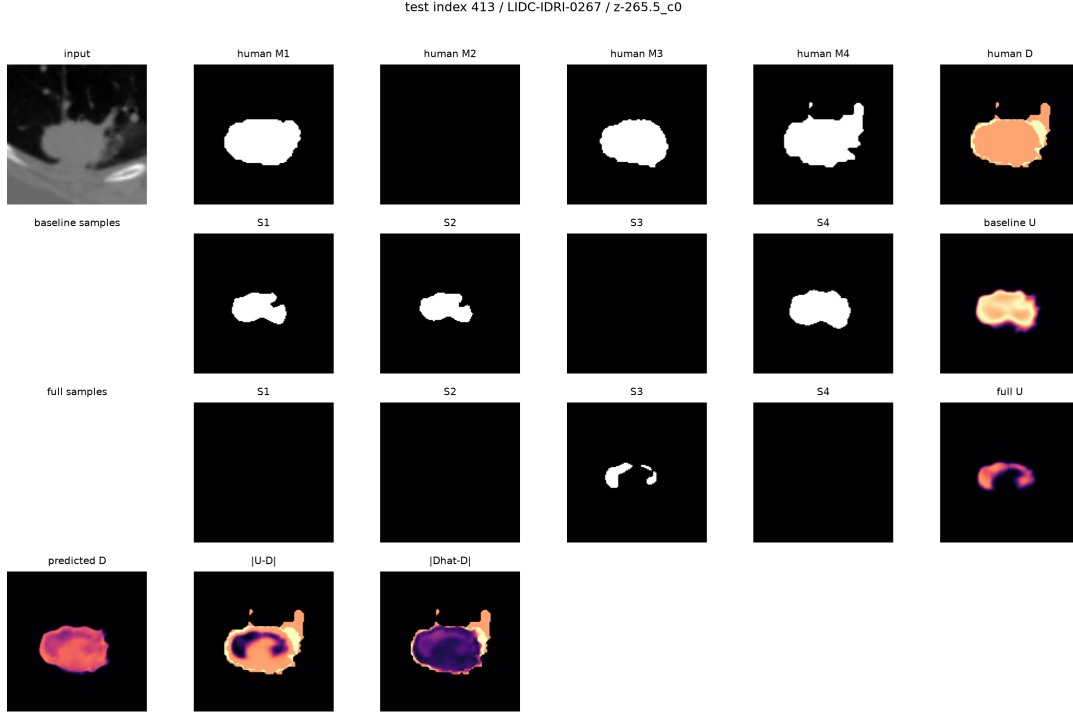


Figure 2: A case in which the graders disagree about whether a lesion exists at all (test index 413, LIDC-IDRI-0267). Row 1: the CT crop, the four expert masks — note that M2 is empty — and D^{GT} , which covers the whole nodule rather than a rim, because the disagreement concerns existence. Rows 2–3: four samples and the resulting U for the baseline and for our full model, both reproducing the ambiguity by emitting a mixture of empty and non-empty samples. Row 4: the predicted $\hat{D}$ and the two error maps. $|\hat{D} - D^{\text{GT}}|$ is dark over the lesion while $|U - D^{\text{GT}}|$ is bright, which illustrates the limitation discussed below: the head recovers existence-disagreement, but U would have to be maximal across the entire nodule to match D^{GT} , and that competes directly with the segmentation loss.

of the loss: the best point estimate in the study and no distribution at all.

The sharpest limitation is the marginal itself. D^{GT} is a per-pixel quantity, so two graders drawing mask A and two drawing mask B yield exactly the same map as four graders disagreeing incoherently; neither D^{GT} nor U can distinguish coherent modes from pixel noise, which bounds both the supervision and the evaluation. Beyond that, we train for 100k of the paper’s 240k iterations with crop-only augmentation, so our absolute energy distances sit above published values; we study one dataset and one anatomy; D^{GT} is estimated from only four graders; and each arm was run with a single seed, so the Dice gap of arm B cannot be separated from run-to-run variation.

Three directions follow. Restricting L_{align} to a boundary band would let it shape edge placement without demanding uncertainty across whole lesions, and is in our view the most likely route to a decisive Q2 result. Arm D might be rescued by adding an explicit diversity or repulsion term, or by retaining the KL alongside the matching loss, so that the objective has some pressure keeping samples apart. Finally, comparing against the Hierarchical Probabilistic U-Net and against MoSE on the same split would place these numbers in context.

References

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